

TarangMini ST22LR01 Code Loading User Guide

Document Version: 01

Release Date: 27-08-2021

Download and install the STM32 CubeProgrammer application software

This section provides instructions for downloading and installing the STM32 CubeProgrammer. If the TarangMini ST22LR01 module is not automatically detected when it is attached to your computer, you may also need to install the USB drivers.

Requirements

To program TarangMini ST22LR01 module with STM32 CubeProgrammer application software, you must connect the **TarangMini demo board** [USB adapter board] to your computer. Programming requires below piece of external hardware:

- USB A-Mini Cable

Install STM32 CubeProgrammer application software

1. Navigate to <https://www.st.com/en/development-tools/stm32cubeprog.html#overview&secondary=st-news-highlights>
2. Select the latest version of the software from the dropdown list and click on the Get software to download file.



3. When the file finishes downloading, run the executable file and follow the steps in the STM32 CubeProgrammer setup Wizard.



Install USB drivers

The required USB drivers are automatically installed the first time you connect a TarangMini Demo board to your computer. If the board drivers do not automatically install, use the following instructions to install the board drivers manually.

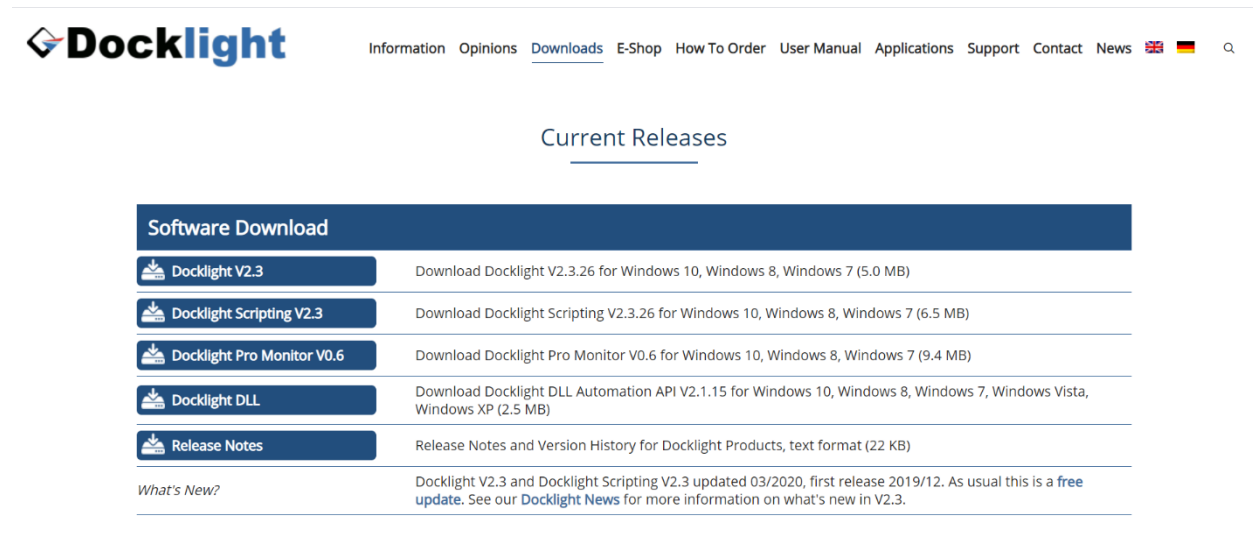
1. Go to the FTDI drivers page: <https://www.ftdichip.com/Drivers/VCP.htm>
2. Locate the correct driver for your operating system
3. For the Windows operating systems, click the **setup executable** link. A zip file Downloads.

Operating System	Release Date	Processor Architecture							Comments
		x86 (32-bit)	x64 (64-bit)	PPC	ARM	MIPSII	MIPSIV	SH4	
Windows*	2017-08-30	2.12.28	2.12.28	-	-	-	-	-	WHQL Certified. Includes VCP and D2XX. Available as a setup executable . Please read the Release Notes and Installation Guides.
Linux	-	-	-	-	-	-	-	-	All FTDI devices now supported in Ubuntu 11.10, kernel 3.0.0-19 Refer to TN-101 if you need a custom VCP VID/PID in Linux. VCP drivers are integrated into the kernel.
Mac OS X 10.3 to 10.8	2012-08-10	2.2.18	2.2.18	2.2.18	-	-	-	-	Refer to TN-105 if you need a custom VCP VID/PID in MAC OS
Mac OS X 10.9 and above	2019-12-24	-	2.4.2	-	-	-	-	-	This driver is signed by Apple
Windows CE 4.2-5.2**	2012-01-06	1.1.0.20	-	-	1.1.0.20	1.1.0.10	1.1.0.10	1.1.0.10	
Windows CE 6.0/7.0	2016-11-03	1.1.0.22 CE 6.0 CAT CE 7.0 CAT	-	-	1.1.0.22 CE 6.0 CAT CE 7.0 CAT	1.1.0.10	1.1.0.10	1.1.0.10	For use of the CAT files supplied for ARM and x86 builds refer to AN_319
Windows CE 2013	2015-03-06	1.0.0			1.0.0				VCP Driver Support for WinCE2013

4. Right click the zip and select Extract All. A folder displays with the setup file
5. Double click the setup file to run it
6. Follow the steps in the installation wizard.

Install Docklight Tool

1. Navigate to <https://docklight.de/downloads/>
2. Select the latest and compatible version of the software for your PC



Docklight Information Opinions Downloads E-Shop How To Order User Manual Applications Support Contact News   

Current Releases

Software Download	
 Docklight V2.3	Download Docklight V2.3.26 for Windows 10, Windows 8, Windows 7 (5.0 MB)
 Docklight Scripting V2.3	Download Docklight Scripting V2.3.26 for Windows 10, Windows 8, Windows 7 (6.5 MB)
 Docklight Pro Monitor V0.6	Download Docklight Pro Monitor V0.6 for Windows 10, Windows 8, Windows 7 (9.4 MB)
 Docklight DLL	Download Docklight DLL Automation API V2.1.15 for Windows 10, Windows 8, Windows 7, Windows Vista, Windows XP (2.5 MB)
 Release Notes	Release Notes and Version History for Docklight Products, text format (22 KB)
<i>What's New?</i>	Docklight V2.3 and Docklight Scripting V2.3 updated 03/2020, first release 2019/12. As usual this is a free update . See our Docklight News for more information on what's new in V2.3.

3. When the file finishes downloading, run the executable file and follow the steps in the Docklight setup Wizard.

Hardware overview

This section provides information about the steps required to work with TarangMini Demo board [USB adapter board].

General features

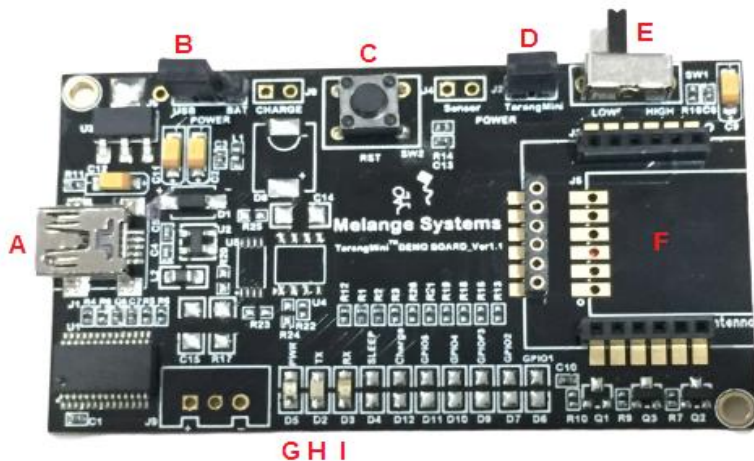
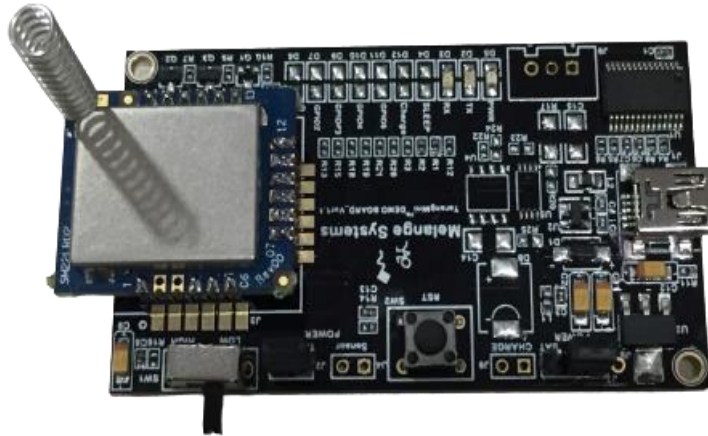


Image: TarangMini Demo board [USB adapter board]

- A. USB Mini Connector
- B. Jumper- USB Mode
- C. Reset Button
- D. Jumper- TarangMini Mode
- E. Jumper- Boot Mode
- F. TarangMini ST22LR01 Module socket connector
- G. Status LED- Power
- H. Status LED- TXD
- I. Status LED- RXD

Steps to be followed for TarangMini Demo Board,

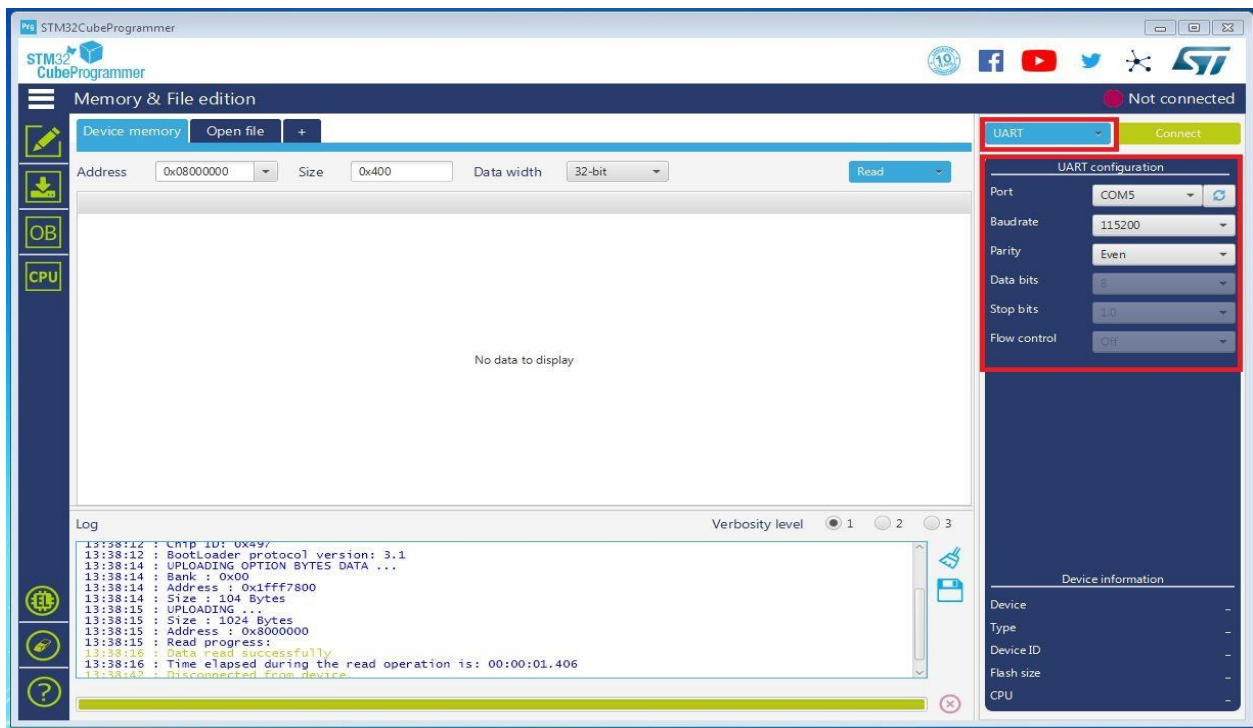
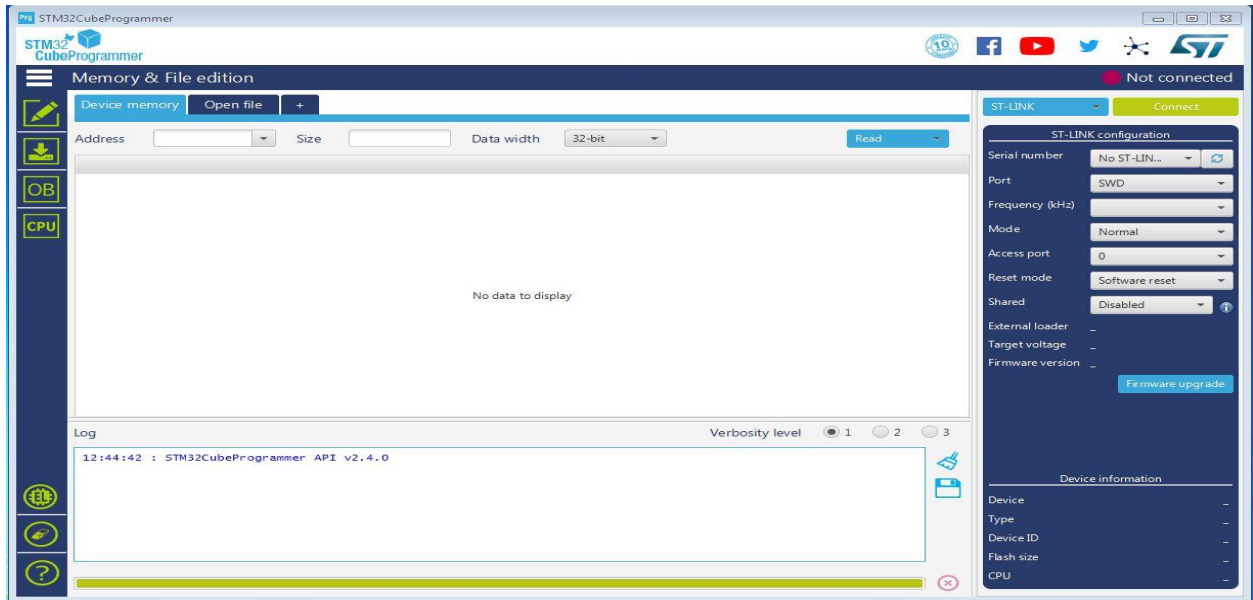
- 1> The TarangMini ST22LR01 module must be in a TarangMini demo board [USB adapter board] connected to your PC with the USB cable.



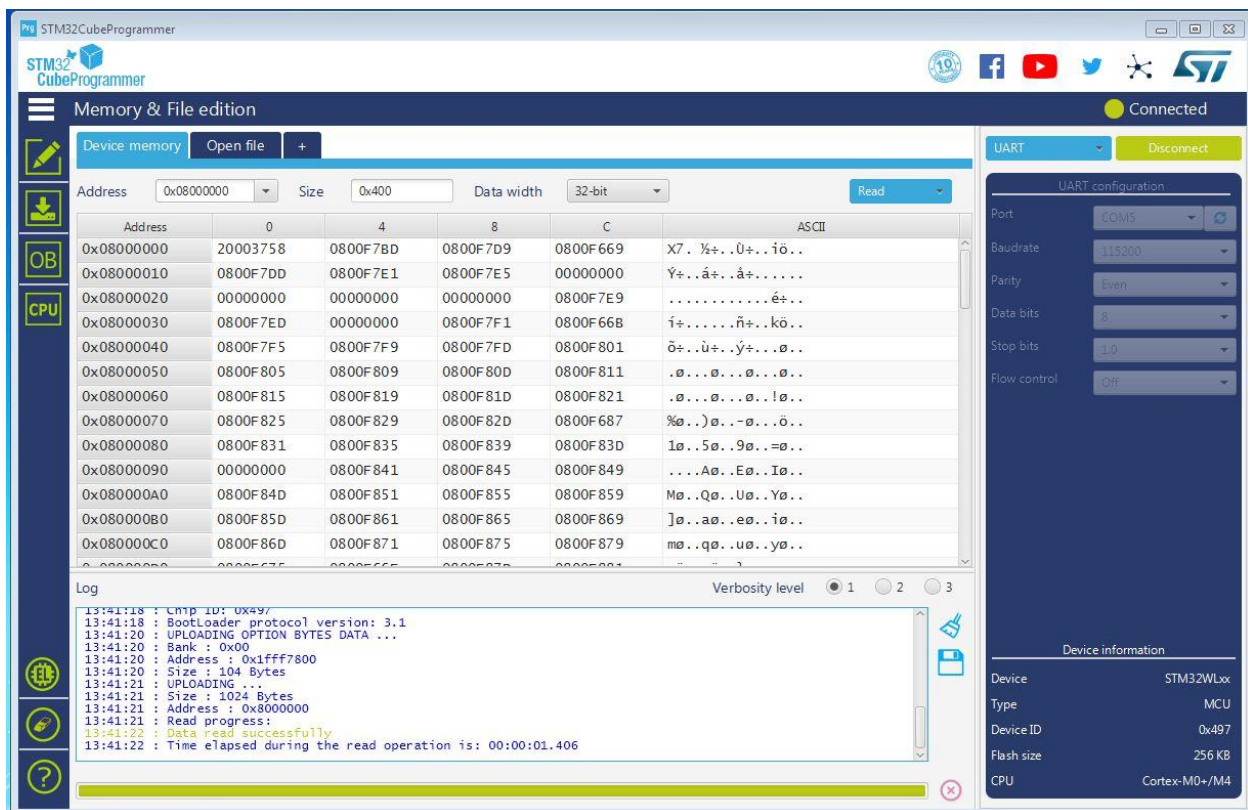
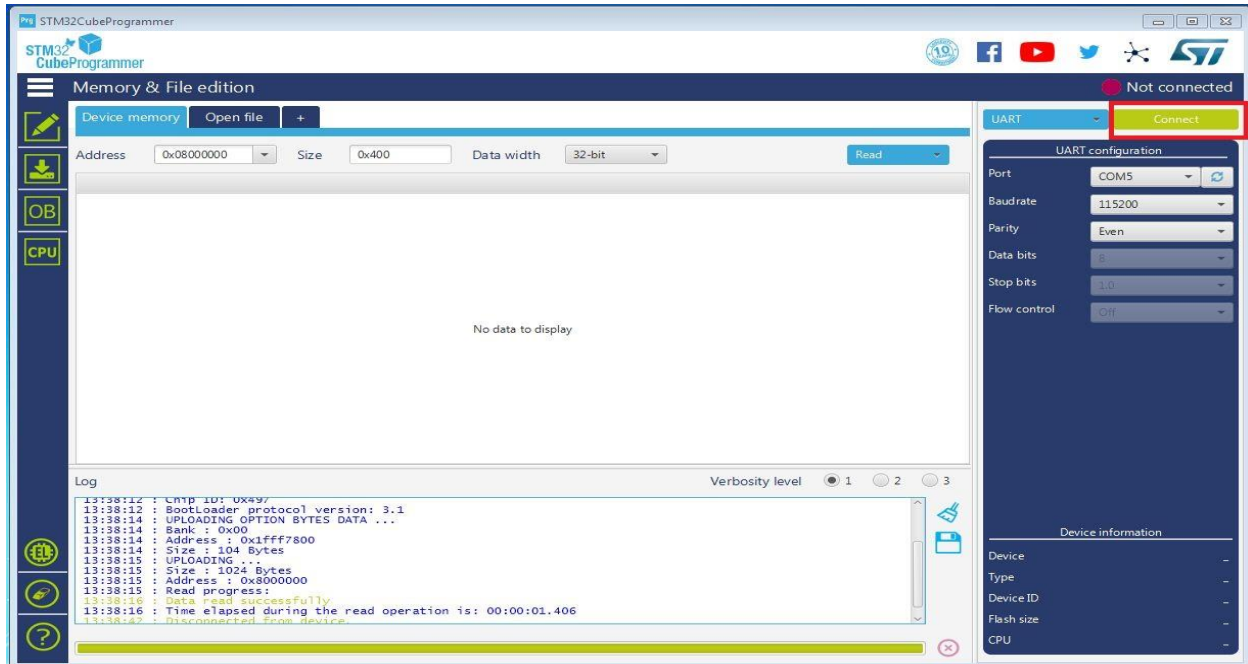
- 2> Insert the TarangMini ST22LR01 module
- 3> Select the Boot switch to **HIGH**
- 4> Apply reset on the board

Steps to be followed on the STM32 CubeProgrammer,

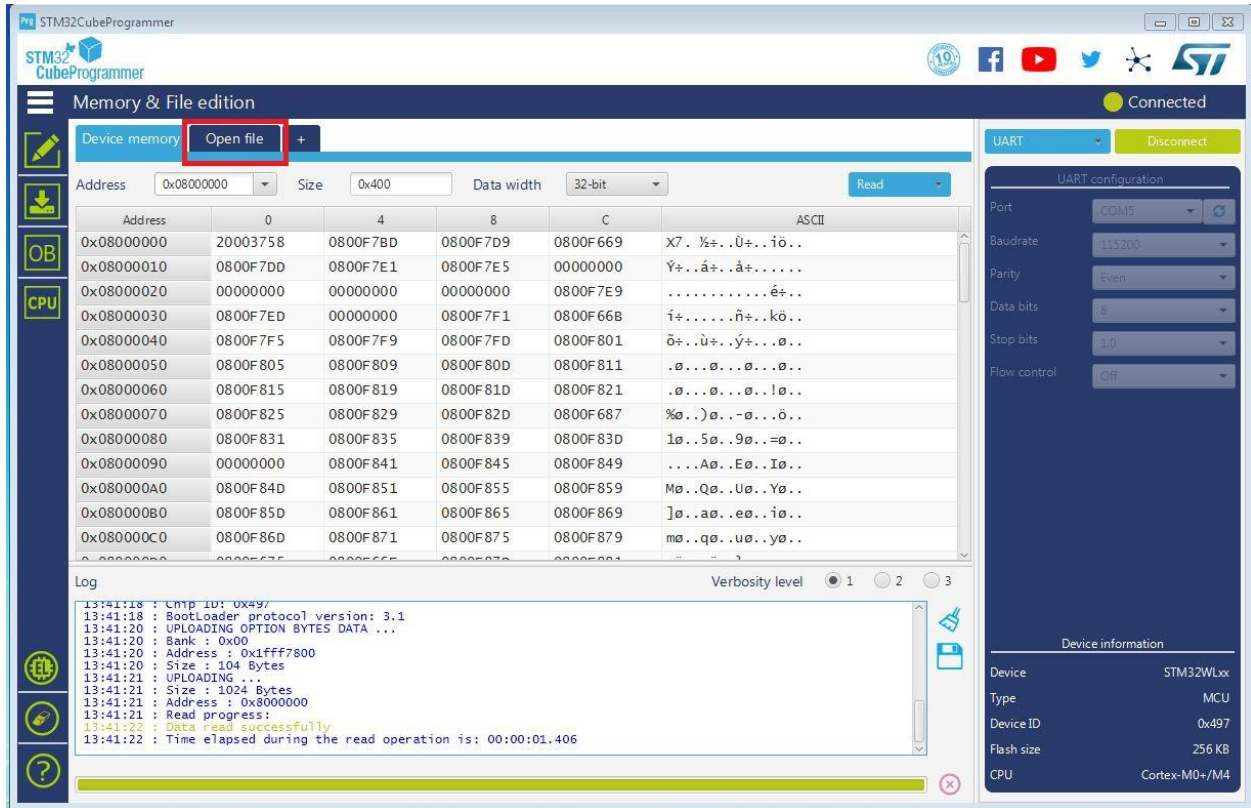
1> Select the COM port



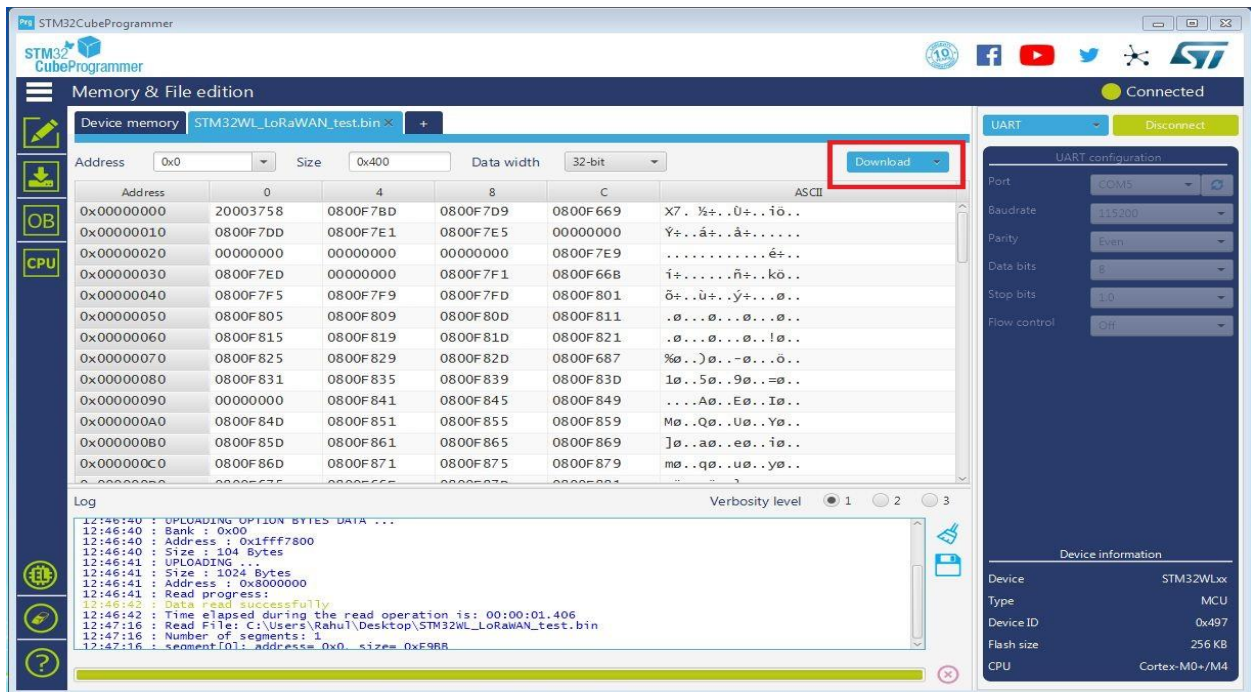
2> Click to connect



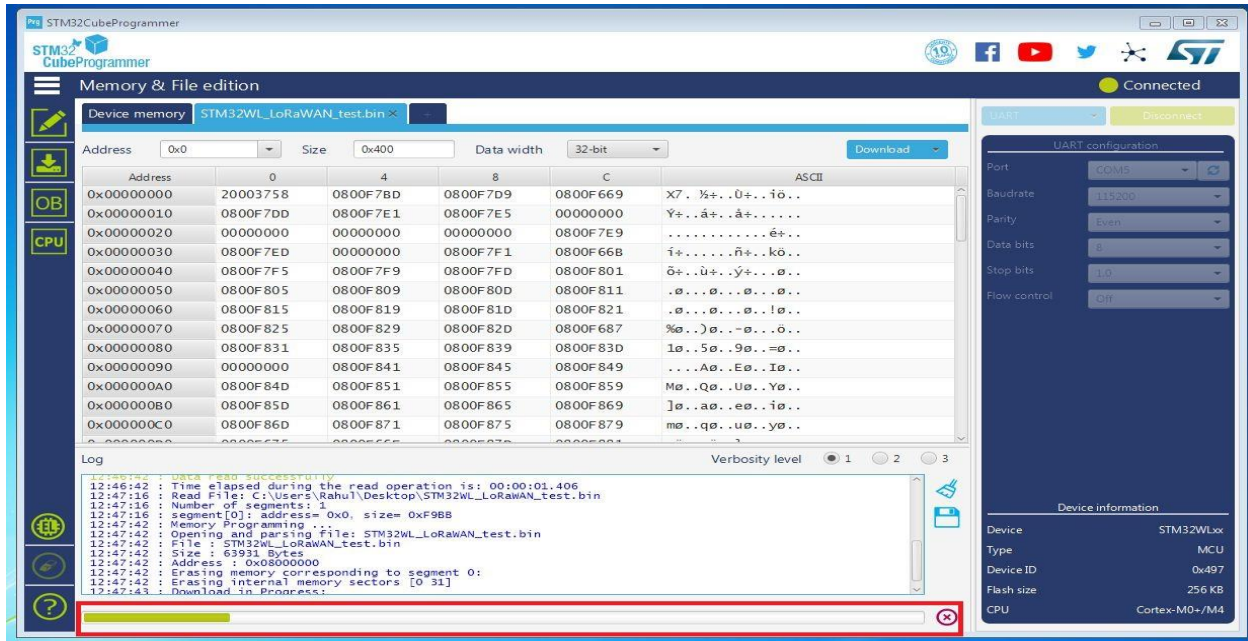
3> Open file for selecting the binary file



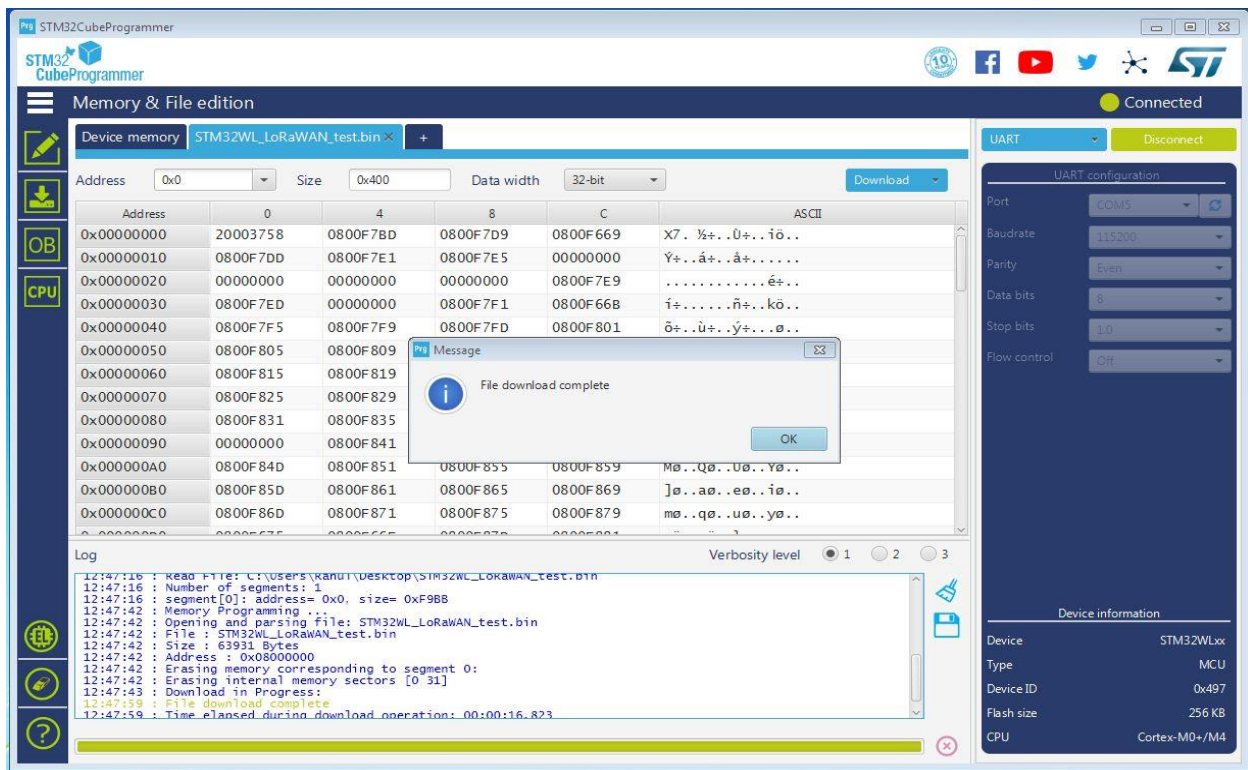
4> Click on the download button to flash the selected binary file



5> Downloading in process...



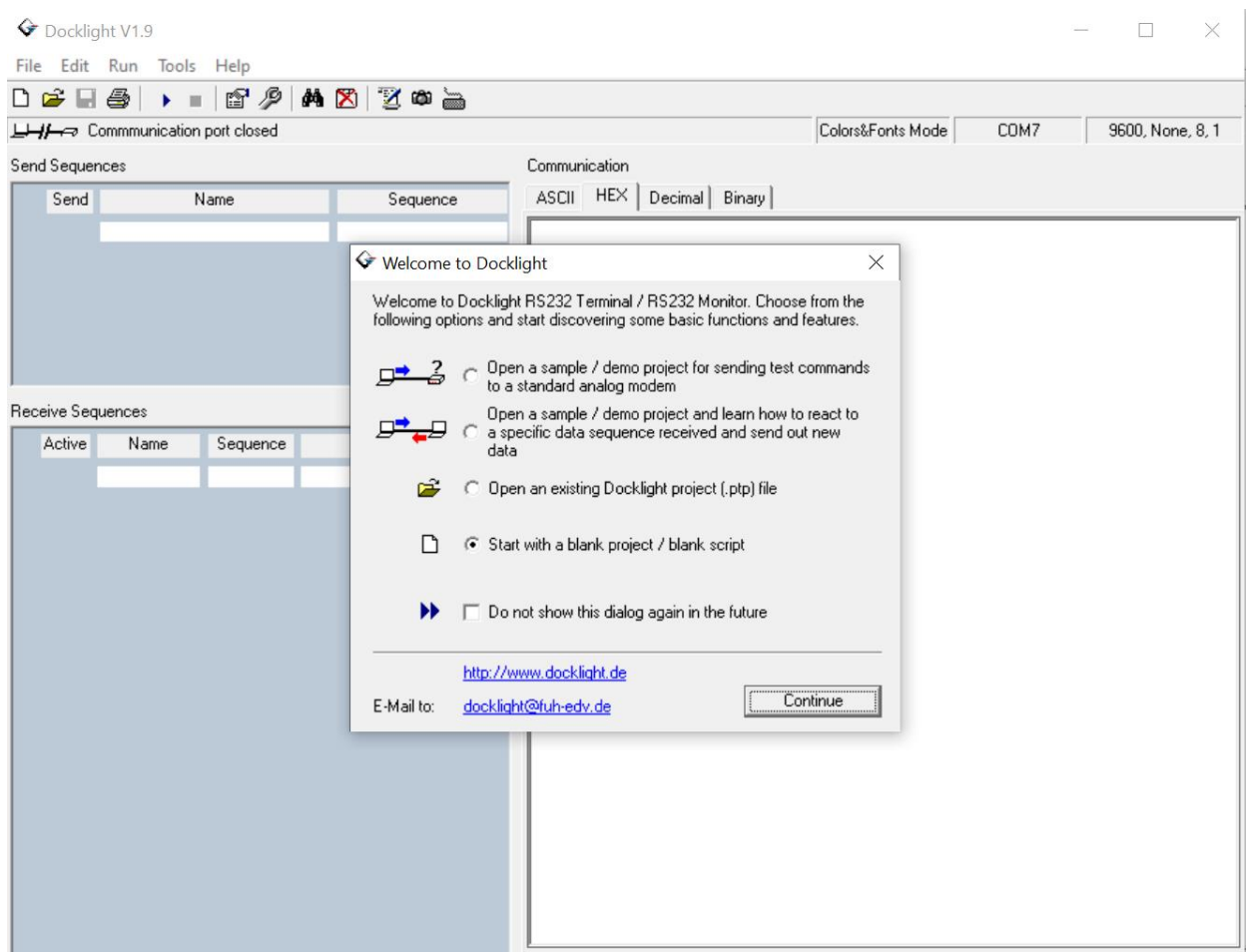
6> Upgraded Successfully



- 7> Click to **Disconnect** the STMCubeProgrammer tool
- 8> Select the Boot switch to **LOW**
- 9> Apply reset on the board
- 10> Issue the API command to check the version of firmware loaded by using the docklight tool (or) any other serial terminal tool as given below.

Steps to be followed on the docklight tool

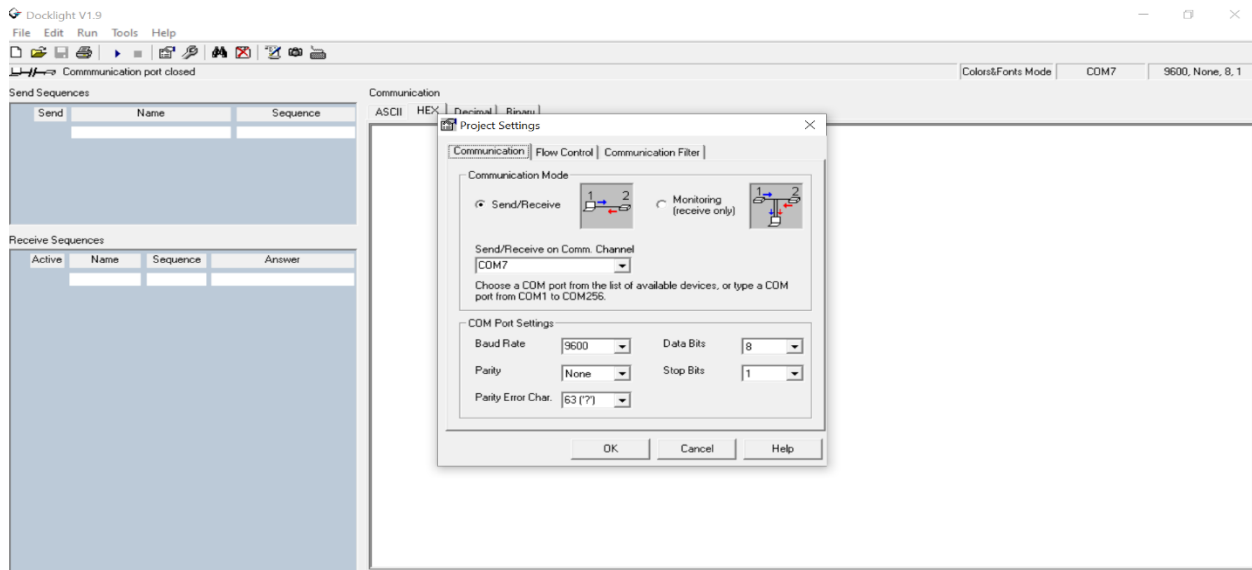
- 1> Create a new project



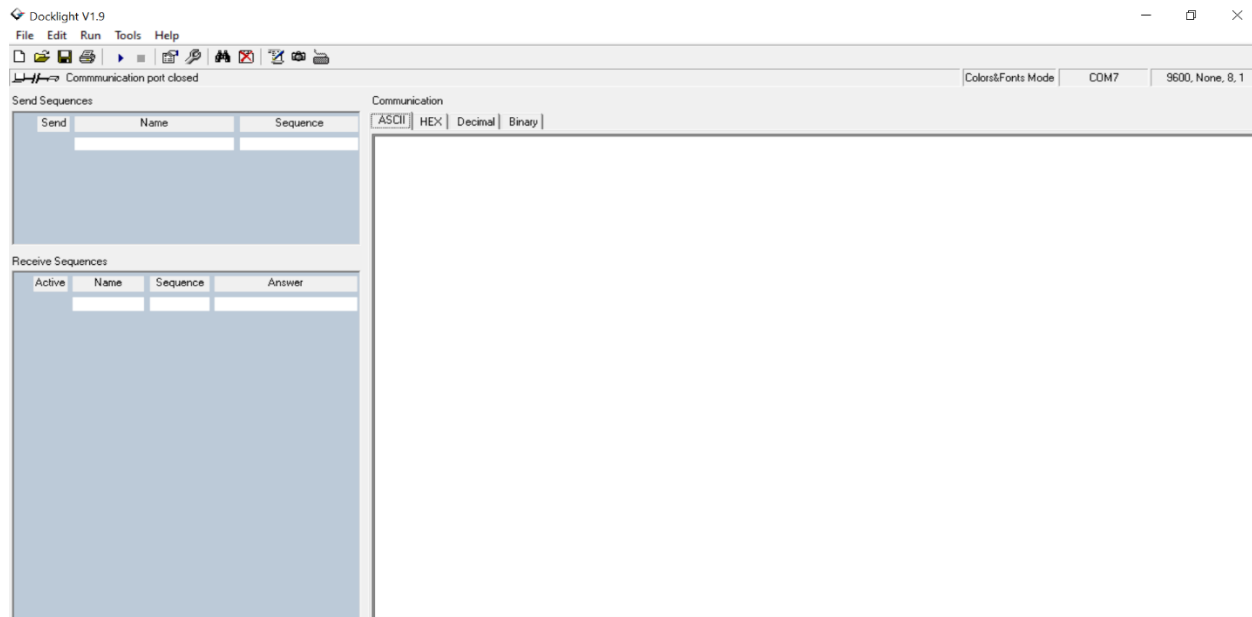
- 2> Click to **Continue**

3> Setting the communication options

- Choose the menu **Tools > Project Settings...**
- Choose communication mode **Send/Receive**
- At **Send/Receive on comm.** channel, set the COM Port where your serial device is connected.
- Set the baud rate and all other **COM Port Settings** required.

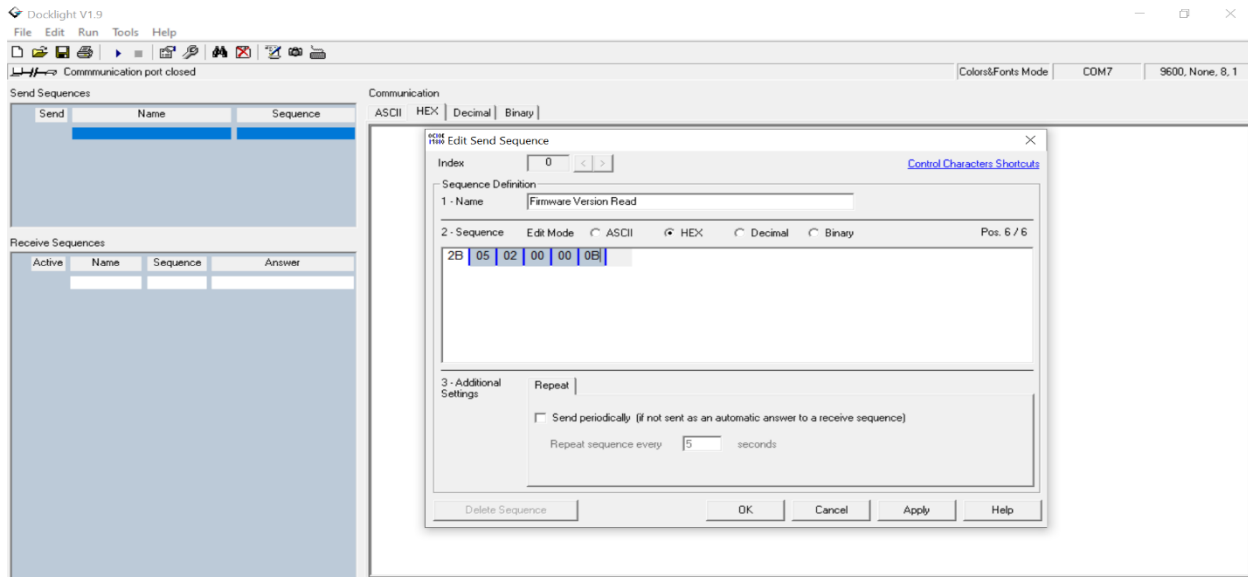


- Confirm the settings and close the dialog by clicking the **OK** button.

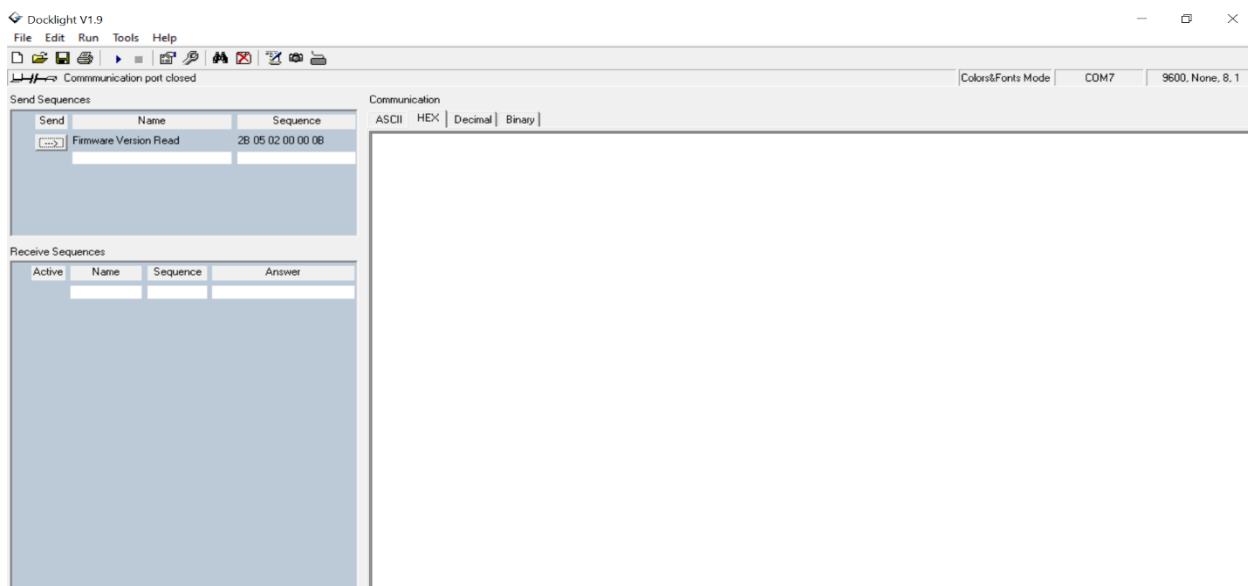


4> Defining the send sequence

- Double click on the last line of the **Send Sequences** table. The **Edit Send Sequence** dialog is displayed.
- Enter a **Name** for the sequence. The sequence name should be unique for every Send Sequence defined.
- Enter the **Sequence** itself. You may enter the sequence in **HEX** format.



- After clicking the **OK** button, the new sequence will be added to the Send Sequence lists



5> Running the test

- Start Docklight by choosing **Run > Start communication**
- Docklight will open a serial connection according to the parameters specified. It will then display all incoming and outgoing communication in the communication window. Use the ---> **Send** Button to send one of the defined sequences to the serial device.

